

Theorem 1.1: formal statement and initial verified lemmas

Falconer distance-gauge formalization project
October 8, 2026

Status. This is an incomplete formalization notebook. The main theorem is an explicit proof obligation; it has not passed Comparator and is not registered on Palomar. The results below record the proved foundation.

The exact target

The uploaded manuscript defines

$$h_\theta(r) = r \exp\left(-(\log(1/r))^\theta\right) \quad (0 < r < 1), \quad h_\theta(0) = 0.$$

Its Theorem 1.1 asserts that, for a compact set $E \subseteq \mathbb{R}^2$,

$$\frac{2}{3} < \theta < 1, \quad \mathcal{H}^{h_\theta}(E) > 0 \quad \implies \quad \mathcal{L}^1(\Delta(E)) > 0.$$

$$\Delta(E) = \{\|x - y\| : x, y \in E\}.$$

The formalization uses the Euclidean norm and Mathlib's arbitrary-gauge Hausdorff measure constructor. For radii at least one, the gauge is extended by the identity; this does not affect its small-radius Hausdorff measure. The covering characterization has been proved:

$$\mathcal{H}^{h_\theta}(E) = \sup_{\rho > 0} \inf_{\substack{E \subseteq \bigcup_{n \geq 0} U_n \\ \text{diam}(U_n) \leq \rho}} \sum_{\substack{n \geq 0 \\ U_n \neq \emptyset}} h_\theta(\text{diam}(U_n)).$$

The independent `Challenge.lean` imports only Mathlib and repeats all custom definitions. The solution does not import Challenge. The theorem statement has no auxiliary Frostman, projection, or energy hypothesis.

The geometric foundation

Proposition 1. *For $0 < r < 1$, the gauge is positive and satisfies $h_\theta(r) \leq r$. Consequently $\mathcal{H}^{h_\theta}(E) \leq \mathcal{H}^1(E)$, and countable sets have zero gauge measure. Positive gauge measure supplies two distinct points of E .*

Proof. The exponential is strictly positive. The logarithm and its real power are nonnegative, so the exponential is at most one. Mathlib's gauge comparison theorem gives the measure inequality. Singletons are null for $\mathcal{H}^1$, hence also for the gauge measure; countable subadditivity gives the countable case. A set containing at most one point is countable. $\square$

The all-pairs distance map is continuous. It therefore maps the compact product $E \times E$ to a compact, Lebesgue-measurable distance set. Two distinct points give a positive distance; that fact alone does not give positive length.

The scalar summability foundation

Proposition 2. *For every $s, c > 0$, one has $\log n \leq cn^s$ for all sufficiently large natural numbers n . For every $\theta, c > 0$,*

$$\sum_{n=0}^{\infty} \exp(-cn^{\theta}) < \infty.$$

Proof. Mathlib proves $\log x = o(x^s)$ as $x \rightarrow \infty$ for $s > 0$. Restricting its eventual norm bound to natural numbers proves the first assertion. Applying that assertion with coefficient $c/2$ gives

$$\log n \leq \frac{c}{2}n^{\theta} \implies \exp(-cn^{\theta}) \leq \exp(-2 \log n) = n^{-2}.$$

The inverse-square series converges. The omitted finite initial segment has no effect on summability, so comparison proves the claim. $\square$

Proposition 3. *The branch-counting exponent and filter-error power obey*

$$3\theta - 2 > 0 \iff \theta > \frac{2}{3}, \quad \theta > \frac{2}{3} \implies -8(2\theta - 1) < -1.$$

Consequently, in the theorem's parameter range, every positive coefficient eventually absorbs the logarithm into $n^{3\theta-2}$, and

$$\sum_{n=0}^{\infty} n^{-8(2\theta-1)} < \infty.$$

Proof. The exponent inequalities are linear arithmetic. Apply the logarithmic domination result with exponent $3\theta - 2$, and apply Mathlib's real-power series criterion to the filter-error exponent. $\square$

These results justify scalar convergence once suitable shell energies and filter errors are supplied. They do not establish the geometric estimates that produce those bounds.

Remaining analytic work and verification

The proof still needs the gauge Frostman construction, logarithmically weighted Fourier and projection estimates, endpoint Orlicz radial projection control, the summable Fourier reconstruction criterion, quantitative regular decomposition, the four uniform energy estimates, and the multiscale budget induction. Fixed-parameter estimates require new uniform bounds before they can be used with parameters that shrink as the terminal scale grows.

The main theorem currently has one explicit proof hole. Comparator permits the three standard axioms and must reject the target's proof hole. A successful ordinary build of this skeleton is not a successful theorem verification.

The public project includes a separate sandboxed Comparator job and a manual full Palomar preflight. Registration requires a completed proof, verification of the exact public commit, truthful metadata, and the registry review.

[Public project: CoolRmal/falconer-theta-gauge](#)